

An Order-94 Rowley Template and the Bound

$R_5(5) \geq 42078$

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23 July 2026

Abstract

We give an explicit effective Rowley $(5, 5, 3)$ template of order 94 with extension parameter $\phi = 40$. Its two inherited distance colours remain K_5 -free under period-93 repetition through the exact compression cutoff $4(94 - 2) = 368$, while its third distance class is a sum-free triangle-free template. We combine this template with an explicit cyclic three-colouring of K_{453} recovered from Rowley’s archived ancillary data. The resulting five distance classes partition $\{1, \dots, 42076\}$ and define a cyclic five-colouring of K_{42077} . A self-contained specialization of Rowley’s composition argument proves that none of the five colours contains K_5 . Consequently,

$$R_5(5) \geq 42078.$$

All finite claims are checked by exact algorithms. The order-94 template and order-453 prototype are each accepted by independent Python and C++ clique searches using different internal representations, and the full compound word is reconstructed independently from the displayed set-union formula. The checked revision of the standard survey records the previous bound $R_5(5) \geq 41626$; priority beyond the literature corpus described here remains subject to external review.

Keywords. Multicolour Ramsey numbers; linear colourings; circulant colourings; triangle-free templates; exact computation.

MSC 2020. 05C55, 05D10, 68W30.

1 Introduction

For positive integers r and k , the diagonal multicolour Ramsey number $R_r(k)$ is the least N such that every r -colouring of the edges of K_N contains a monochromatic K_k . An explicit r -colouring of K_{N-1} without monochromatic K_k proves $R_r(k) \geq N$.

Rowley developed a composition method in which a small linear colouring containing a distinguished triangle-free template is combined with a second linear Ramsey colouring [1, 2]. In the case relevant

*Draft for independent review. Mathematical correctness, priority, and publication readiness are treated as separate questions throughout.

here, his published order-93 effective $(5, 5, 3)$ template, with $\phi = 40$, combines with an order-453 three-colouring to give a five-colouring of order 41625, and hence

$$R_5(5) \geq 41626.$$

This value is recorded in revision 18 of Radziszowski's survey [3].

The contribution of this paper is one additional template position. We exhibit an order-94 effective $(5, 5, 3)$ template with the same value $\phi = 40$. The compound order becomes

$$(94 - 1)(453 - 1) + 1 + 40 = 93 \cdot 452 + 41 = 42077.$$

Thus the one-position extension is amplified into a gain of 452 vertices.

The proof has three logically separate inputs.

- Section 3 gives the explicit order-94 word and the exact finite conditions it satisfies.
- Section 4 gives the explicit order-453 prototype recovered from Rowley's archived 2022 ancillary data.
- Sections 5 and 6 define the compound and prove directly that its five colours are K_5 -free.

Search heuristics are not part of the proof. The frozen words, the composition formula, and exact checkers form the certificate.

Remark 1.1 (Status and scope). The theorem proved here is a correctness claim. The specific value 42078 was not found in the checked literature, and the checked survey revision records 41626, but those facts do not establish priority. A public priority claim should await an independent literature search and contact with specialists familiar with unpublished linear Ramsey constructions.

2 Linear colourings and triangle-free templates

Definition 2.1 (Linear distance colouring). For $m \geq 2$, a linear q -colouring of K_m is a map

$$c : \{1, \dots, m - 1\} \longrightarrow \{1, \dots, q\}.$$

The edge $\{i, j\}$, where $0 \leq i < j \leq m - 1$, receives colour $c(j - i)$. It is cyclic if $c(d) = c(m - d)$ for every $1 \leq d \leq m - 1$.

For a three-colour word c , put

$$A_1 = c^{-1}(1), \quad A_2 = c^{-1}(2), \quad T = c^{-1}(3).$$

Definition 2.2 (Triangle-free template). The third distance class T is a triangle-free template, or *tf-template*, if $m - 1 \in T$ and the colour-3 graph is triangle-free. Its extension parameter is

$$\phi = \min(T) - 1.$$

Lemma 2.3 (Sum-free criterion). *The colour-3 graph of a linear word is triangle-free if and only if there do not exist positive integers a, b such that*

$$a \in T, \quad b \in T, \quad a + b \in T.$$

The case $a = b$ is included.

Proof. Translate the least vertex of any triangle to 0 and write the other two vertices as $a < a+b$. Its three edge lengths are a , b , and $a+b$. The converse is immediate from the same three vertices. $\square$

For a period $p = m - 1$, the inherited colours are repeated by assigning a positive distance d the base colour

$$c(((d-1) \bmod p) + 1). \quad (1)$$

Thus residue zero is represented by base distance p , not by 0. This convention is essential because the terminal distance p belongs to the template colour.

Definition 2.4 (Effective $(5, 5, 3)$ certificate). An order- m word is an effective $(5, 5, 3)$ certificate with parameter ϕ if:

1. $m - 1 \in T$ and $\min(T) - 1 = \phi$;
2. T satisfies Lemma 2.3; and
3. for each $s \in \{1, 2\}$, the periodic colour defined by (1) has no K_5 on vertices

$$0, 1, \dots, 4(m-2).$$

The final cutoff is the specialization to K_5 of the compression argument in Rowley’s Generalised Construction Theorem [1, Definition 3.1 and Theorem 3.2]. For completeness, the same compression is written explicitly in Section 6.

3 The order-94 template

Certificate 3.1 (Canonical order-94 word). Let $c(1)c(2)\cdots c(93)$ be the following word; displayed line breaks are typographic only:

1112122122221211121221211121222122121113131232132323333333333333333333
333333323312321313

Its distance classes are

$$\begin{aligned} A_1 &= \{1, 2, 3, 5, 8, 13, 15, 16, 17, 19, 22, 24, 25, 26, 28, 33, 36, 38, 39, 40, \\ &\quad 42, 44, 48, 86, 90, 92\}, \\ A_2 &= \{4, 6, 7, 9, 10, 11, 12, 14, 18, 20, 21, 23, 27, 29, 30, 31, 32, 34, 35, 37, \\ &\quad 45, 47, 50, 52, 82, 84, 87, 89\}, \\ T &= \{41, 43, 46, 49, 51, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, \\ &\quad 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 83, 85, 88, 91, 93\}. \end{aligned}$$

Proposition 3.2 (Exact finite properties). *Certificate 3.1 is an effective $(5, 5, 3)$ certificate of order 94 with $\phi = 40$. More explicitly:*

1. $93 \in T$, $T \cap \{1, \dots, 40\} = \emptyset$, and $\min(T) = 41$;
2. no $a, b, a + b$ all lie in T , including the case $a = b$;
3. neither inherited colour contains K_5 on vertices $0, \dots, 368$; and

4. the reflection identities

$$c(d) = c(41 - d) \quad (1 \leq d \leq 40), \quad c(d) = c(134 - d) \quad (41 \leq d \leq 93)$$

hold.

Proof. Items (1), (2), and (4) are direct complete enumerations on the displayed word. For item (3), two independent exact clique searches construct the periodic colour graphs on the 369 vertices $0, \dots, 368$ and exhaustively reject K_5 in both inherited colours. The Python checker uses arbitrary precision integer bit masks for forward-neighbour sets; the C++ checker uses ordered integer candidate vectors and pairwise filtering. Their algorithms and regression coverage are described in Section 7. $\square$

The canonical file is `results/order94_t12.template`, with SHA-256

`a3dd3415956277d68fc0197f6f5e35c38f2ed7936c66f11f658b0c57688abec1.`

The same word is also accepted when overtested through distance 372. Two further distinct accepted order-94 words are retained in the artifact package as robustness evidence, but they are not used below.

4 The order-453 prototype

Rowley's 2022 construction starts from a cyclic order- n prototype, copies its colours at distances $1, \dots, n-1$, and places the new template colour at distance n [2, Section 5]. The archived order-977 (5, 5, 5, 3) template in ancillary sheet `Paper_Sep_2022`, column AB, was obtained by extending an Exoo-derived order-453 (5, 5, 5) graph. Consequently, entries $1, \dots, 452$ of that column are the prototype word, and entry 453 begins colour 4.

The full 452-symbol word is printed in Appendix A.

Proposition 4.1 (Source-recovered prototype). *The word in Appendix A defines a cyclic three-colouring of K_{453} with no monochromatic K_5 .*

Proof. Every symbol was compared with the SHA-256-fixed ancillary XML. The comparison also checks the sheet metadata and confirms that distance 453 begins the new template colour. Independent Python bit-mask and C++ ordered-vector searches then exhaustively reject K_5 in each of the three finite colour graphs. Finally, direct comparison gives $v(d) = v(453-d)$ for all $1 \leq d \leq 452$. $\square$

The extracted artifact is `sources/rowley_exoo_order453.prototype`, with SHA-256

`19c97e6279c184f6f462786cadda4b7c9773d870a5b680a04eb1503ef384a2d0.`

Once the word is explicit and Proposition 4.1 is verified, the historical attribution is not a logical dependency of the main theorem.

5 The compound colouring

Set $p = 93$. Let A_1, A_2, T be the distance classes in Certificate 3.1, and let B_1, B_2, B_3 be the three distance classes of the order-453 prototype.

For $s \in \{1, 2\}$, define the inherited output classes

$$A'_s = \{\ell + (\mu - 1)p : \ell \in A_s, 1 \leq \mu \leq 452\} \cup \{\ell + 452p : \ell \in A_s, 1 \leq \ell \leq 40\}. \quad (2)$$

For $j \in \{1, 2, 3\}$, relabel the three prototype colours as output colours $j + 2$ and put

$$B'_j = \{\ell + (\mu - 1)p : \ell \in T, \mu \in B_j\}. \quad (3)$$

Lemma 5.1 (Partition). *The five sets $A'_1, A'_2, B'_1, B'_2, B'_3$ partition $\{1, \dots, 42076\}$.*

Proof. First consider $1 \leq d \leq 452p$. Write d uniquely as $d = \ell + (\mu - 1)p$ with $1 \leq \ell \leq p$ and $1 \leq \mu \leq 452$. If $\ell \in A_s$, then $d \in A'_s$. If $\ell \in T$, then μ lies in exactly one prototype class B_j , and $d \in B'_j$. These cases are disjoint and exhaustive.

The remaining distances are $452p + \ell$ with $1 \leq \ell \leq 40$. Since $T \cap \{1, \dots, 40\} = \emptyset$, each such ℓ lies in exactly one of A_1, A_2 , and the second union in (2) assigns the distance uniquely. $\square$

Colour the edge $\{x, y\}$, where $0 \leq x < y \leq 42076$, by the unique one of the five sets in Lemma 5.1 containing $y - x$. This is the complete explicit colouring.

Table 1: Distance-class sizes in the compound colouring.

Output colour	Number of distances	Origin
1	11772	first inherited template colour
2	12676	second inherited template colour
3	5304	prototype colour 1
4	7566	prototype colour 2
5	4758	prototype colour 3
Total	42076	all positive distances

6 Proof of the Ramsey bound

Theorem 6.1. *There is a five-colouring of K_{42077} with no monochromatic K_5 . Consequently,*

$$R_5(5) \geq 42078.$$

Proof. We separately exclude cliques in inherited colours and in prototype-derived colours.

Inherited colours. Fix $s \in \{1, 2\}$ and suppose that a colour- s K_5 exists in the compound. Translate its least vertex to 0 and write the other vertices as

$$0 < h_1 < h_2 < h_3 < h_4.$$

Every h_i and every difference $h_j - h_i$ has a residue in A_s modulo $p = 93$.

If $h_1 > p$, subtract p from all four positive coordinates. If a consecutive gap $h_{i+1} - h_i$ exceeds p , subtract p from the entire upper tail $h_{i+1}, \dots, h_4$. These operations preserve positivity, strict order, and every relevant colour: differences across the cut change by exactly p , while differences within either part do not change.

Iteration terminates with $h_1 \leq p$ and every consecutive gap at most p . No inherited-colour edge can have length divisible by p , because the representative of residue zero is $p = 93 \in T$. Hence $h_1 \leq p - 1$, every consecutive gap is at most $p - 1$, and

$$h_4 \leq 4(p - 1) = 368.$$

This gives a K_5 in the exact periodic graph rejected by Proposition 3.2, a contradiction.

Prototype-derived colours. Fix $j \in \{1, 2, 3\}$ and suppose that output colour $j + 2$ contains a K_5 . Translate its least vertex to 0. Each other vertex has a unique form

$$d_i = t_i + (\mu_i - 1)p, \quad t_i \in T, \quad \mu_i \in B_j.$$

Take two vertices with $d_i < d_k$. If $t_i < t_k$, then the positive residue of $d_k - d_i$ is $t_k - t_i$. This residue cannot lie in T , because

$$t_i + (t_k - t_i) = t_k$$

would contradict the sum-free property of T . Thus an edge in the same prototype-derived colour forces $t_i \geq t_k$.

If $t_i = t_k$, the difference has prototype index $\mu_k - \mu_i$. If $t_i > t_k$, then

$$d_k - d_i = (p - (t_i - t_k)) + (\mu_k - \mu_i - 1)p,$$

whose prototype index is again $\mu_k - \mu_i$. Membership of every pairwise difference in output colour $j + 2$ therefore implies

$$\mu_k - \mu_i \in B_j.$$

The indices μ_i are strictly increasing because the d_i increase while the t_i do not. Hence

$$\{0, \mu_1, \mu_2, \mu_3, \mu_4\}$$

is a monochromatic K_5 in prototype colour j , contradicting Proposition 4.1.

All five output colours are therefore K_5 -free. The compound has order

$$93 \cdot 452 + 41 = 42077,$$

so the definition of $R_5(5)$ gives $R_5(5) \geq 42078$. □

Corollary 6.2. *The linear colouring in Theorem 6.1 is cyclic.*

Proof. The prototype is cyclic, and Certificate 3.1 satisfies Rowley's two reflection identities stated in Proposition 3.2(4). Direct comparison of the expanded word also gives $w(d) = w(42077 - d)$ for every $1 \leq d \leq 42076$. □

7 Exact computational verification

The computational claims in Propositions 3.2 and 4.1 are finite, exhaustive, and deterministic. No random search decision enters their verdicts.

7.1 Clique search

Both implementations use the following exhaustive recursion, with different representations of the candidate set.

```
Search(prefix, candidates, need):
  if need = 0: return prefix
  if size(candidates) < need: return NONE
  while size(candidates) >= need:
    v = least vertex in candidates
    delete v from candidates
    next = {u in candidates : Adj(v,u)}
    result = Search(prefix followed by v, next, need-1)
    if result is not NONE: return result
  return NONE
```

Every candidate list is strictly increasing. At each call it consists exactly of the larger vertices adjacent to every prefix vertex. Thus every clique is represented once. The cardinality test prunes only branches that cannot reach the target size, so returning **NONE** is an exhaustive proof of nonexistence for the supplied finite graph.

7.2 Independent implementations and finite counts

The Python implementation stores forward-neighbour sets in arbitrary precision integer bit masks. The C++ implementation stores ordered integer vectors and forms intersections by explicit pairwise filtering. They share the written mathematical semantics but neither parser nor clique-search code.

Table 2: Exact search-node counts. Different counts reflect different candidate orderings and data structures.

Finite object and checker	Colour 1	Colour 2	Colour 3
Order-94 template, Python	550261	718490	–
Order-453 prototype, Python	1040265	5811421	782585
Order-453 prototype, C++	237855	837235	201199

Seven negative fixtures test malformed length, a missing terminal template distance, forbidden-prefix use, an additive template triple, insufficient repeat span, and explicit repeated K_5 witnesses in each inherited colour. Both template implementations make the same accept/reject decisions on the complete fixture suite.

7.3 Source extraction and compound reconstruction

The order-93 published seed and the order-453 prototype are compared symbol for symbol with the ancillary XML in the fixed arXiv source archive. The archive SHA-256 is

5041096a33d6898c116a35a102e521fb14fe872fb18c20ff7d30822ed4c396b2.

The final expanded word is generated once by a block/residue implementation of (2) and (3). A second program, which does not import the generator, constructs the five displayed sets as literal unions, verifies that every distance $1, \dots, 42076$ occurs exactly once, compares every symbol with the frozen word, checks cyclic reflection, and confirms SHA-256

a97d5dce8a927db2889ba220119f9d4f3d1b88ee24d441f82a803a195ae8028d.

Remark 7.1. The 42077-vertex graph was not subjected to a naive all-vertex clique search. Its K_5 -freeness is established by the two preservation arguments in Theorem 6.1, whose finite hypotheses are checked exactly.

8 Reproducibility and review protocol

The complete deterministic verification is run from the artifact package by

```
sh tests/run_end_to_end_checks.sh
```

The suite:

1. re-extracts the published order-93 seed and order-453 prototype from the fixed ancillary XML;
2. verifies all three prototype colours with the Python and C++ clique searches;
3. verifies three distinct order-94 words with the Python and C++ full-condition checkers;
4. overtests the canonical word through distance 372; and
5. reconstructs all 42076 compound distances independently and checks the frozen hash.

An adversarial reproduction should pay particular attention to:

- the representative 93 for residue zero;
- inclusion of $a = b$ in the sum-free test;
- the distinction between maximum distance 368 and vertex count 369;
- the source-theorem wording caveat concerning the aggregate cutoff, versus the explicit per-colour bound in its proof; and
- the recovery of distances $1, \dots, 452$ from ancillary column AB, with distance 453 beginning template colour 4.

The remaining nonmathematical gates are an independent-machine rerun, a clean-room third checker if desired, and a specialist priority search.

results/r5_5_order42077.linear-coloring.

Expanded 42076-symbol five-colour distance word. SHA-256:

a97d5dce8a927db2889ba220119f9d4
f3d1b88ee24d441f82a803a195ae8028d.

sources/arxiv-2203.13476-source.tar.gz.

Fixed source archive containing the ancillary XML. SHA-256:

5041096a33d6898c116a35a102e521fb
14fe872fb18c20ff7d30822ed4c396b2.

References

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